B9.6 Systems with nonreal eigenvalues

So still dealing with system:

$$x'(t) = Ax(t)$$

where

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

and A is an $n \times n$ matrix with real constant entries.

Say A has at least one nonreal eigenvalue (therefore at least two since they come in conjugate pairs), denote them $\alpha \pm \beta i$ for α, β real and $\beta \neq 0$

And for $\alpha + \beta i$ say an associated eigenvector is $a + bi$ where a, b are $n \times 1$ matrices/vectors.

Be careful not to confuse the symbols for the eigenvalues vs. eigenvectors here, much easier to do that here compared to real eigenvalues due to the nature/additional symbols.

It turns out that

$$w(t) = e^{(\alpha + \beta i)t}(a + bi)$$

is a solution to $x'(t) = Ax(t)$, same form as when the eigenvalue was real so quite natural. Why? Observe:

$$w'(t) = (\alpha + \beta i)\, e^{(\alpha + \beta i)t}(a + bi)$$

while

$$Aw(t) = A e^{(\alpha + \beta i)t}(a + bi)$$

$$= (\alpha + \beta i) e^{(\alpha + \beta i)t}(a + bi)$$

Last step being because $a + bi$ is an eigenvector for A.

So indeed $w(t)$ does work as we said. Ok this is a bit dicey since for $w'(t)$ we used

$$(e^{\lambda t})' = \lambda e^{\lambda t}$$

which is of course true if λ is real. But as it isn't here, is that ok? Turns out it is, can break it down via Euler's identity for e^{it} if nothing else.

So $w(t)$ is a solution for the desired system. But not really the solution we want as it generally is not real.

So consider

$$w(t) = w_1(t) + w_2(t)\, i$$

where $w_1(t), w_2(t)$ are real functions.

Then since we showed $w'(t) = Aw(t)$ that means

$$[\, w_1(t) + w_2(t)\, i\,]' = A\,[\, w_1(t) + w_2(t)\, i\,]$$

so equating real and imaginary parts on both sides, since A is real, that yields:

$$w'_1(t) = Aw_1(t)$$
$$w'_2(t) = Aw_2(t)$$

So since $w_1(t), w_2(t)$ are real, they are really the solutions that we want. And this helps validate why differentiating with the i makes sense as it being just a constant multiple of real function doesn't seem atypical.

How to express these solutions conveniently? Note

$$w(t) = e^{(\alpha + \beta i)t}(a + bi)$$

$$= e^{\alpha t} \cdot e^{i \cdot \beta t}(a + bi)$$

$$= e^{\alpha t}[\cos(\beta t) + i \sin(\beta t)](a + bi)$$

So then multiplying this out and collecting the real and imaginary parts gives us

$$w_1(t) = e^{\alpha t}[\cos(\beta t)a - \sin(\beta t)b]$$

$$w_2(t) = e^{\alpha t}[\sin(\beta t)a + \cos(\beta t)b]$$

so these are the solutions corresponding to the eigenvalue $\alpha + \beta i$

Note be careful about sign of terms in these, they are not fluid / interchangeable (common mistake)!

E.g. w_1 is real part of w, flipping sign of one term it won't be that anymore nor a constant multiple of it. So can't do that.

We seem to have two solutions for the one eigenvalue $\alpha + \beta i$ and that doesn't seem natural as usually each eigenvalue of algebraic multiplicity 1 ought to have only one associated solution?

Well what would be the solutions for the conjugate eigenvalue $\alpha - \beta i$?

Since this has conjugated eigenvector, a - bi, also, recreating w_1 , w_2 for this eigenvalue, since cos is an even function and sin is an odd function, in w_1 first term stays same and both pieces of second term flip sign so overall stays same.

Meanwhile in w_2 both terms will flip sign, so will just be the negative of what we had earlier. But since in homogeneous system, constant multiple of solution is still solution, that's equivalent (hence another reason to be careful with signs or that equivalence might not be maintained).

Thus $\alpha - \beta i$ gives same solutions as $\alpha + \beta i$
So really those w_1, w_2 were the solutions for the

pair of eigenvalues: $\alpha \pm \beta i$

And that makes more sense in giving us the proper number of total solutions.

B9.6.5 Solve system

$$x_1' = -x_1 - 2x_2$$
$$x_2' = 8x_1 - x_2$$

Then we have coefficient matrix as:

$$A = \begin{bmatrix} -1 & -2 \\ 8 & -1 \end{bmatrix}$$

So we find eigenvalues of A:

$$A - \lambda I = \begin{bmatrix} -1 - \lambda & -2 \\ 8 & -1 - \lambda \end{bmatrix}$$

meaning:

$$\det(A - \lambda I) = (-1 - \lambda)^2 + 25 \stackrel{?}{=} 0$$

---> $$-1 - \lambda = \pm 4i$$

---> $$\lambda = -1 \pm 4i$$

Consider $\lambda = -1 + 4i$:

$$A - (-1+4i)I = \begin{bmatrix} -4i & -2 \\ 8 & -4i \end{bmatrix}$$

Null space of this? Well really only one equation since this is eigenvalue, so just look at equation created by the first row:

$$(-4i)y_1 - 2y_2 = 0$$

So can just pick say: $y_1 = -1$ ----> $y_2 = 2i$

Remember any nonzero multiple of this also works (real or nonreal) if you picked so that y_2 was real and y_1 was nonreal that wouldn't inherently mean you did anything wrong.

Hence our chosen eigenvector here is:

$$\begin{bmatrix} -1 \\ 2i \end{bmatrix} = \underbrace{\begin{bmatrix} -1 \\ 0 \end{bmatrix}}_{a} + \underbrace{\begin{bmatrix} 0 \\ 2 \end{bmatrix}}_{b} i$$

Thus using prior formulas we have

$$x(t) = c_1 \left\{ e^{-t} \cos(4t) \begin{bmatrix} -1 \\ 0 \end{bmatrix} - e^{-t} \sin(4t) \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right\} +$$

$$c_2 \left\{ e^{-t} \sin(4t) \begin{bmatrix} -1 \\ 0 \end{bmatrix} + e^{-t} \cos(4t) \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right\}$$

So using $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$, we get

$$x_1(t) = -c_1 e^{-t} \cos(4t) - c_2 e^{-t} \sin(4t)$$

$$x_2(t) = 2c_2 e^{-t} \cos(4t) - 2c_1 e^{-t} \sin(4t)$$

What would change if A was 3 x 3 with some nonreal eigenvalues? Well one eigenvalue would be real so so deal with that as we did last time. And the two nonreal, deal with them as we did above.

Just as in diagonalization, of course the calculation will be tougher / longer but same ideas. As usual nonreal eigenvalues tougher in calculation of eigenvector than real eigenvalues but also easier in that in conjugate pair you need only examine one of the two, while distinct real eigenvalues have to each be examined individually.

Even for 10 x 10 same idea, the main thing that might fundamentally cause us problems is repeated eigenvalues (which might be real or nonreal).
as we alluded to in the prior section.

Addendum section will actually clarify this a bit.

Addendum

(1) In a linear homogeneous equation with constant coefficients, when r was repeated root of characteristic equation, we had standard solution of e^{rt} but another solution was te^{rt}, why though?

We can verify it works, especially in second order compared to higher order, via brute forcing it. But what is the logic behind it?

(2) Also in linear, nonhomogeneous with constant coefficients when there was overlap between the homogeneous solution and the function you were trying to create in the nonhomogeneous equation, the key idea was to multiply the usual guess by t, same as above.

But again, why is that the thing that fixes it?

(3) Even in linear homogeneous systems with repeated eigenvalues without the needed number of eigenvectors we used this idea of multiplying by t (Problems B9.5.35/36).

So, in short, that idea of multiplying by t to "fix" anomalous situations is very common, these are three distinct situations which use that idea. But why does it work like that? You've probably just been taking it on faith that it works, no real verification of the idea.

So useful for general understanding to see roughly why that works since it's quite common and because the reason it works is rooted in linear algebra ideas so another crossover between those two pieces of the class.

Ex:

$$y^{(4)} + 2y'' + y = 0$$

Gave this on worksheet. Characteristic Equation is:

$$r^4 + 2r^2 + 1 = 0$$

----> $$(r^2 + 1)^2 = 0$$

----> $$r = i, i, -i, -i$$

Repeated nonreal eigenvalue, quite unusual for us since second order equations can't have those.

So what is general solution? Well $r = i$ gives the solutions $\cos t$, $\sin t$ as usual. But this being fourth order, ought to have four linearly independent solutions?

What two additional solutions do the repetition provide? Perhaps $t \cos t$, $t \sin t$ if we follow the trend? And we can brute force verify that indeed those are the two, but why logically those two?

But other than following the pattern of doing that, we probably don't really have a justification for why we do that specifically.

After all, earlier we did that with e^t style functions $\cos t$, $\sin t$ are quite different, no real assurance will work same way with those.

So again want to clarify this. We will focus on the case of linear, homogeneous, constant coefficient single equation as ideas are similar amongst these contexts.

Say in

$$ay'' + by' + cy = 0$$

that r is a repeated real root of the characteristic equation.

One solution to this is quite easily $y_1 = e^{rt}$ due to r being root of characteristic equation. But what is the other (linearly independent from y_1)?

Ok we've been told what it is, but let's try to find it from scratch. Try variation of parameters (another use of that!). Take:

$$y = u_1(t)\, y_1(t)$$

---->

$$y' = u_1' y_1 + u_1 y_1'$$

---->

$$y'' = u_1'' y_1 + 2u_1' y_1' + u_1 y_1''$$

So plug these into the equation to get:

$$a(u_1''y_1 + 2u_1'y_1' + \underline{u_1y_1''}) + b(u_1'y_1 + \underline{u_1y_1'}) + \underline{cu_1y_1} = 0$$

Since y_1 is a solution, underlined collectively become 0. Inputting that along with $y_1 = e^{rt}$ this becomes:

$$au_1''e^{rt} + 2ar\,u_1'e^{rt} + bru_1'e^{rt} = 0$$

which further simplifies to

$$au_1'' + u_1'(2ar + b) = 0$$

Now since r is repeated root of polynomial $ax^2 + bx + c$ it is also a root of its derivative, $2ax + b$, so that

$$2ar + b = 0$$

meaning above equation becomes:

$$au_1'' = 0$$

thus $u_1'' = 0$ which implies

$$u_1 = Ft + G$$

and hence

$$y = (Ft + G)e^{rt}$$

is the extra solution we were after. The Ge^{rt} was already a solution, so naturally the new part, the te^{rt}, is what we want really.

Hence where that comes from.

Now you might ask why I did this, after all couldn't I have just plugged in te^{rt} and showed it worked?

Yes, I could have but two issues with that.

One, that assumes you know that's the second solution. This method shows how you can get that without having prior knowledge of it ahead of time so that's certainly more instructive.

Second, showing it by brute force plugging it in might be reasonable for an arbtirary second order equation which has r as a repeated root but are you really going to do that with say an arbitrary tenth order equation which has r as a repeated root?

Perhaps the method I showed is more generalizable to higher order? Again interestingly involves linear algebra ideas.

But generalizing this to higher order is a very different thing! Will talk about that next time.